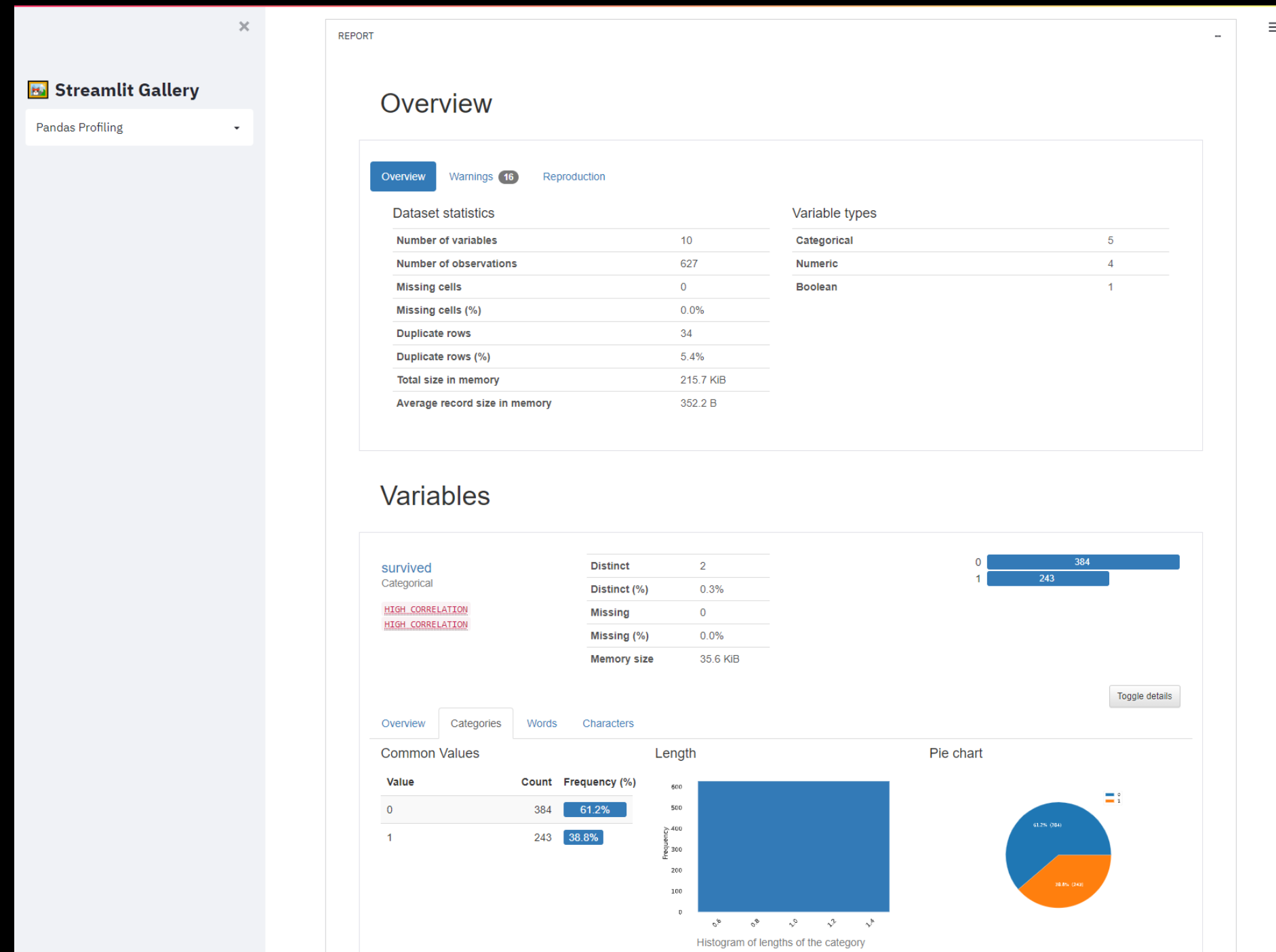


2. Streamlit과 함께하는 데이터 분석

데이터 분석과 머신러닝을 지원하는 툴
데이터 분석 웹 어플리케이션 기능 보유



Rank	Previous	App Store	Genre	Country	U.S.	Views	Global	Views (per 1M user)
1	Sudoku	Angry Birds	Casual	USA	100%	100%	100%	100%
2	Angry Birds POP!	Angry Birds POP!	Casual	USA	100%	100%	100%	100%
3	Angry Birds POP!	Angry Birds POP!	Casual	USA	100%	100%	100%	100%
4	Angry Birds POP!	Angry Birds POP!	Casual	USA	100%	100%	100%	100%
5	Angry Birds POP!	Angry Birds POP!	Casual	USA	100%	100%	100%	100%
6	Angry Birds POP!	Angry Birds POP!	Casual	USA	100%	100%	100%	100%
7	Angry Birds POP!	Angry Birds POP!	Casual	USA	100%	100%	100%	100%
8	Angry Birds POP!	Angry Birds POP!	Casual	USA	100%	100%	100%	100%
9	Angry Birds POP!	Angry Birds POP!	Casual	USA	100%	100%	100%	100%
10	Angry Birds POP!	Angry Birds POP!	Casual	USA	100%	100%	100%	100%

Dataframes

Display a dataframe as an interactive table.

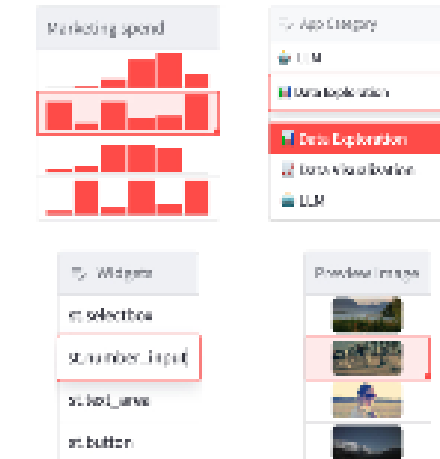
```
st.dataframe(my_data_frame)
```

Name	Age	Gender	Marital Status	Homepage
Emily Chen	35 years	Female	Single	http://www.emily.com
Daniel Martinez	28 years	Male	Single	http://dani.com
Christina Saunders	34	Female	Single	http://pcsa.com
William H. Jones	45 years	Male	Single	http://www.william-h.com
Timothy Barker	32 years	Male	Single	http://thompson.com
Dr. David C. Jones	40 years	Female	Single	http://www.david.com
Alexander	20 years	Male	Single	http://www.alexander.com

Data editor

Display a data editor widget.

```
edited = st.data_editor(df, num_rows="max")
```



Column configuration

Configure the display and editing behavior of dataframes and data editors.

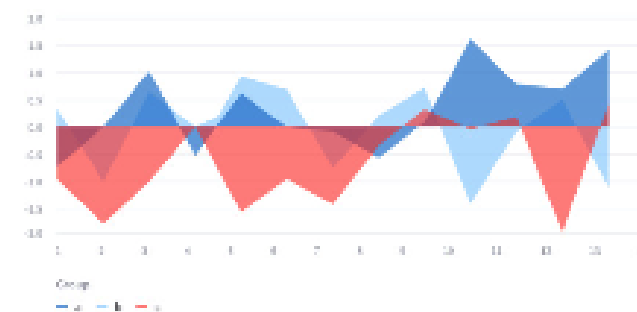
```
st.column_config.NumberColumn(
```



Simple line charts

Display a line chart.

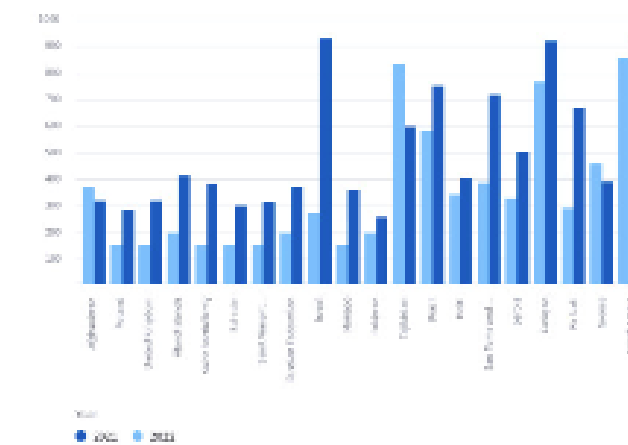
```
st.line_chart(my_data_frame)
```



Simple area charts

Display an area chart.

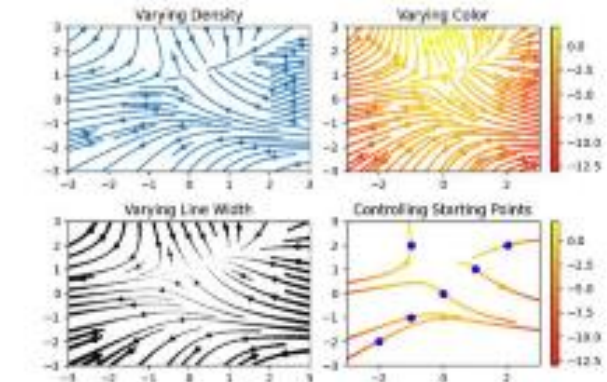
```
st.area_chart(my_data_frame)
```



Simple bar charts

Display a bar chart.

```
st.bar_chart(my_data_frame)
```



Matplotlib

Display a matplotlib.pyplot figure.

```
st.pyplot(my_mpl_figure)
```

Pandas의 데이터 프레임을 렌더링

데이터의 원본 출력 및 확인

출력한 결과 테이블은 사용자와 상호작용 가능

```
import streamlit as st
import pandas as pd
import numpy as np

df = pd.DataFrame(
    np.random.randn(50, 20),
    columns=('col %d' % i for i in range(20)))

st.dataframe(df) # Same as st.write(df)
```

	col 0	col 1	col 2	col 3	col 4	col 5	col 6	col 7	col 8	col 9	col 10
0	-0.1894	0.4283	0.4087	0.3539	0.881	0.392	0.0642	0.0479	0.0957	-0.0703	-0.7732
1	0.5224	0.6675	1.1465	0.4144	0.0334	-0.9711	1.0781	-0.9396	0.1072	-0.0496	-0.2818
2	-0.6847	-0.653	0.15	-0.3329	0.6689	0.2793	-1.3614	1.3484	0.412	0.6549	1.4022
3	-0.3557	-2.3596	-0.8624	0.8539	0.1074	0.7319	-0.1469	0.7568	-0.9802	0.861	0.9945
4	1.5941	0.3398	0.9615	0.8168	-0.8505	0.4503	0.9793	-0.3308	-1.4387	0.6949	0.9555
5	0.4858	0.9	-0.9217	-0.7718	0.404	-0.4346	-0.8378	0.0993	-1.1775	-0.5639	1.8438
6	0.5737	0.1769	-0.6096	2.6857	1.165	-1.1585	1.7803	-0.1448	-1.2124	-1.3247	-1.3514
7	-0.2624	0.7143	2.3	0.7281	-0.7129	-0.416	-1.6043	-0.8089	-0.1224	-1.1428	2.0306
8	0.2879	-0.1469	1.0615	0.2871	0.5955	-1.7389	-1.1996	1.3133	-1.3951	1.6542	-0.079

표에서 **데이터 수정** 가능

수정 결과에 따라 페이지의 다른 내용도
동적으로 반응

사용자와 반응하는 표 삽입

```
import streamlit as st
import pandas as pd

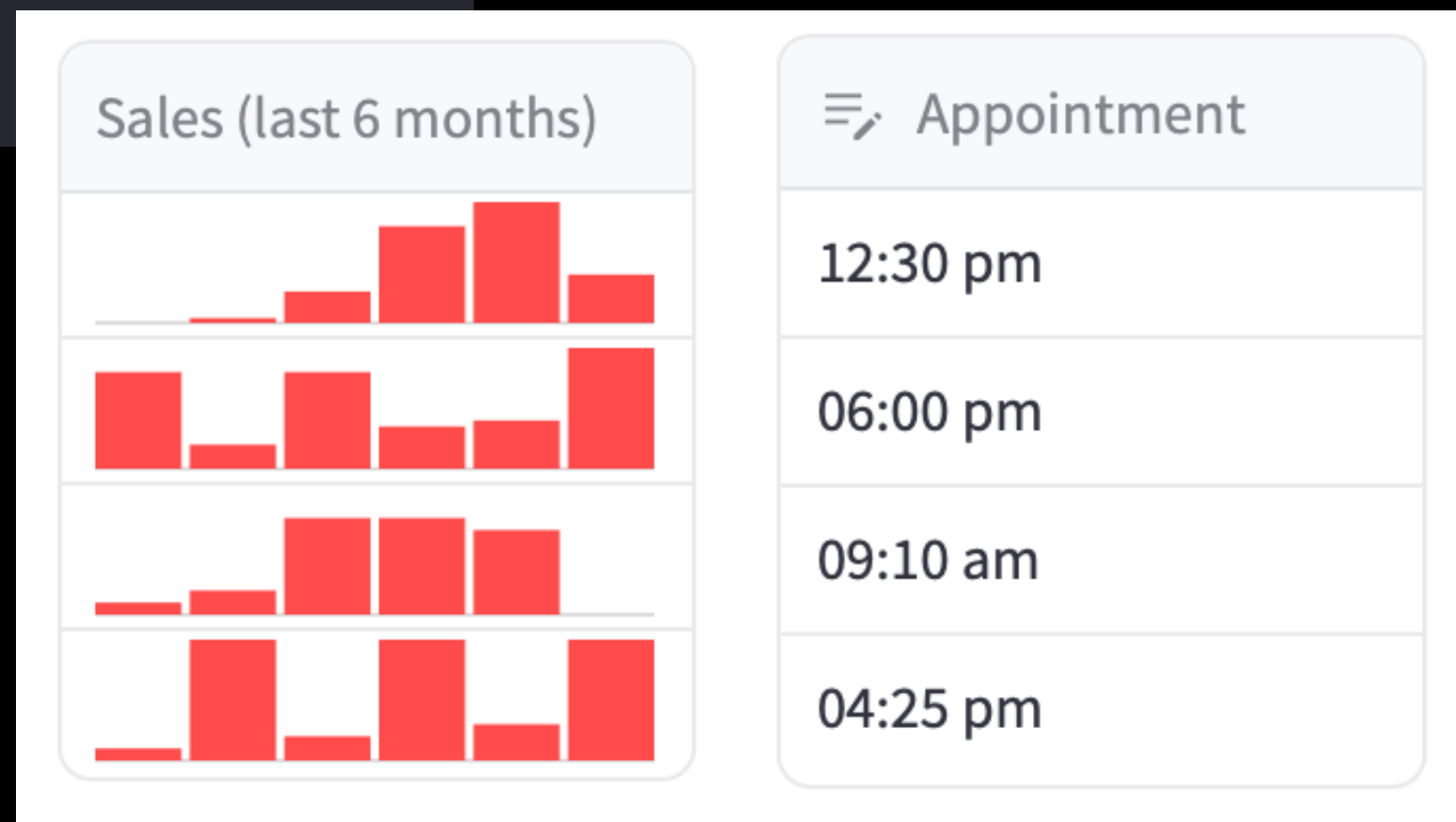
df = pd.DataFrame(
    [
        {"command": "st.selectbox", "rating": 4, "is_widget": True},
        {"command": "st.balloons", "rating": 5, "is_widget": False},
        {"command": "st.time_input", "rating": 3, "is_widget": True},
    ]
)
edited_df = st.data_editor(df)

favorite_command = edited_df.loc[edited_df["rating"].idxmax()][ "command"]
st.markdown(f"Your favorite command is **{favorite_command}** 🍷 ")
```

	command	rating	is_widget
0	st.selectbox	6	☒
1	st.balloons	5	☐
2	st.time_input	3	☒

Your favorite command is st.selectbox 🍷

```
st.data_editor(  
    data_df,  
    column_config={  
        "sales": st.column_config.BarChartColumn(  
            "Sales (last 6 months)",  
            help="The sales volume in the last 6 months",  
            y_min=0,  
            y_max=100,  
        ),  
    },  
    hide_index=True,  
)
```



테이블 안에 **그래프, 체크박스, 이미지**
등을 넣을 수 있도록 돕는 기능

다양한 유형의 표를 넣을 수 있다.



Chart Elements - Line charts

RAG 기반의 챗봇 구현

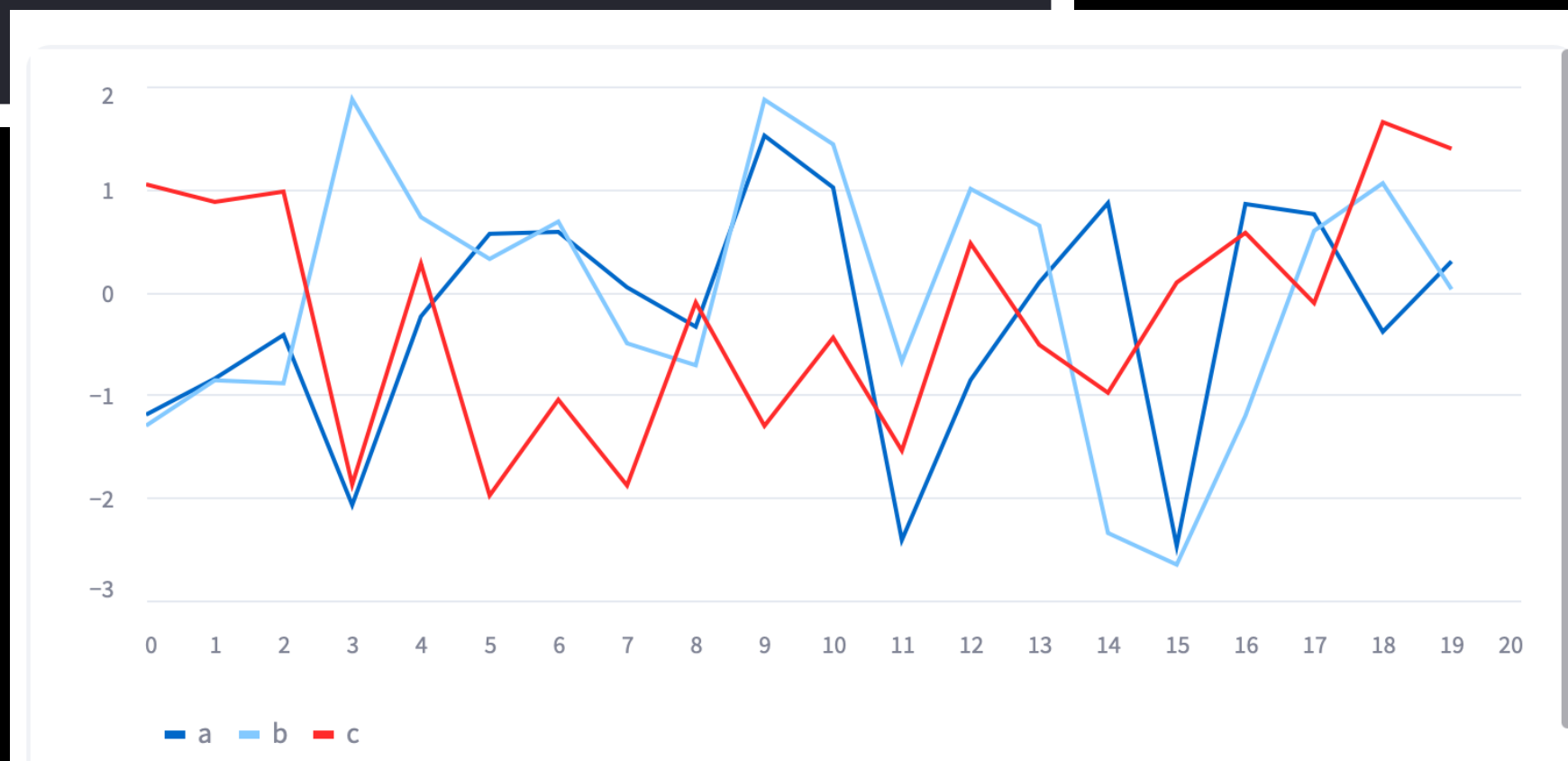
05. Streamlit +
LangChain으로 간단한
웹 어플리케이션 개발

1. 처음 만들어보는
Streamlit 웹 앱 서비스

```
import streamlit as st
import pandas as pd
import numpy as np

chart_data = pd.DataFrame(
    np.random.randn(20, 3),
    columns=['a', 'b', 'c'])

st.line_chart(chart_data)
```



선 그래프를 화면에 출력

연속적으로 변하는 데이터를 한눈에 확인 가능



Chart Elements - Line charts

RAG 기반의 챗봇 구현

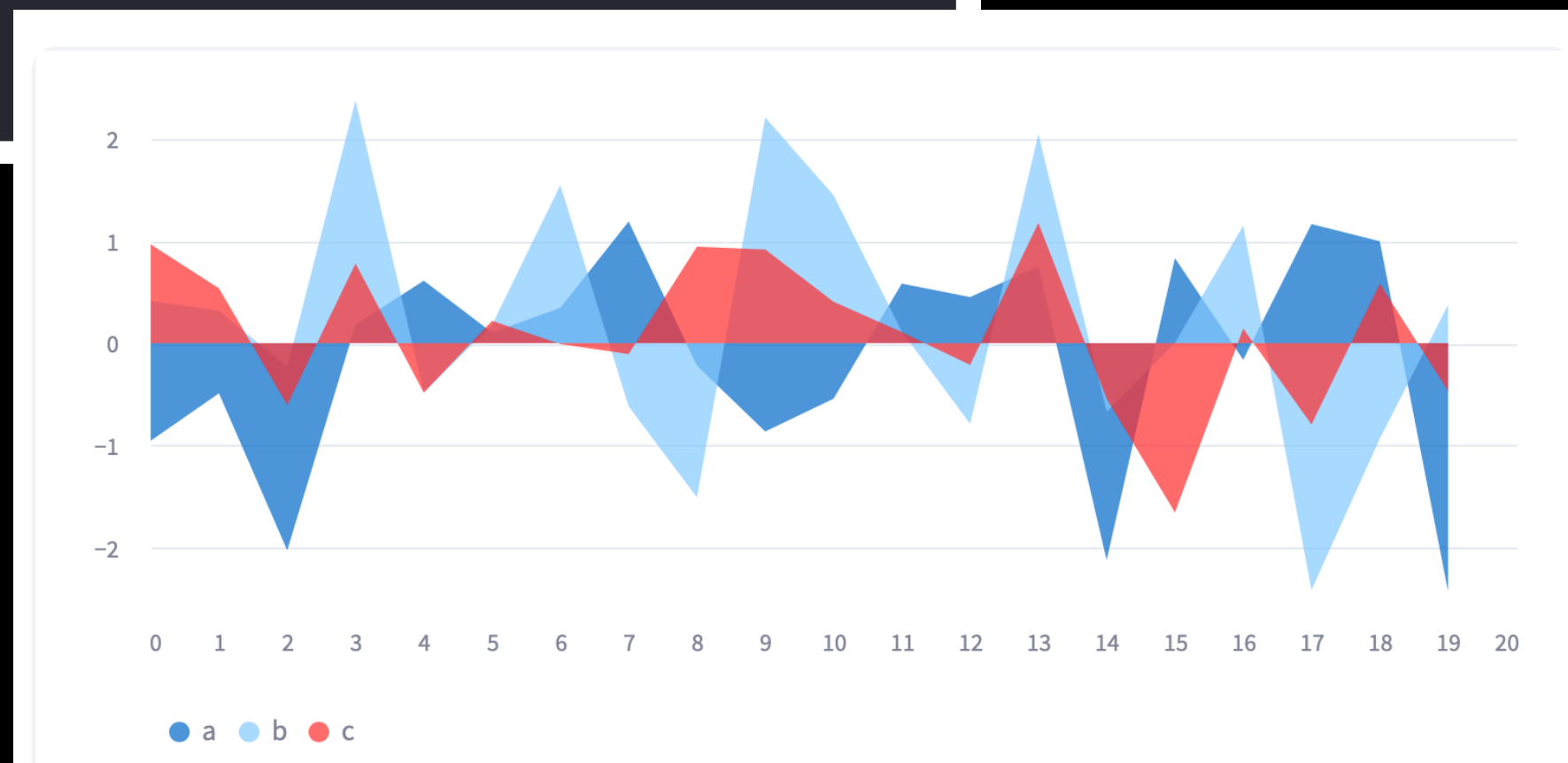
05. Streamlit +
LangChain으로 간단한
웹 어플리케이션 개발

1. 처음 만들어보는
Streamlit 웹 앱 서비스

```
import streamlit as st
import pandas as pd
import numpy as np

chart_data = pd.DataFrame(
    np.random.randn(20, 3),
    columns=['a', 'b', 'c'])

st.area_chart(chart_data)
```



영역 그래프를 화면에 출력

각 데이터가 차지하는 부분을 면적으로
서 확인 가능



Chart Elements – Bar charts

RAG 기반의 챗봇 구현

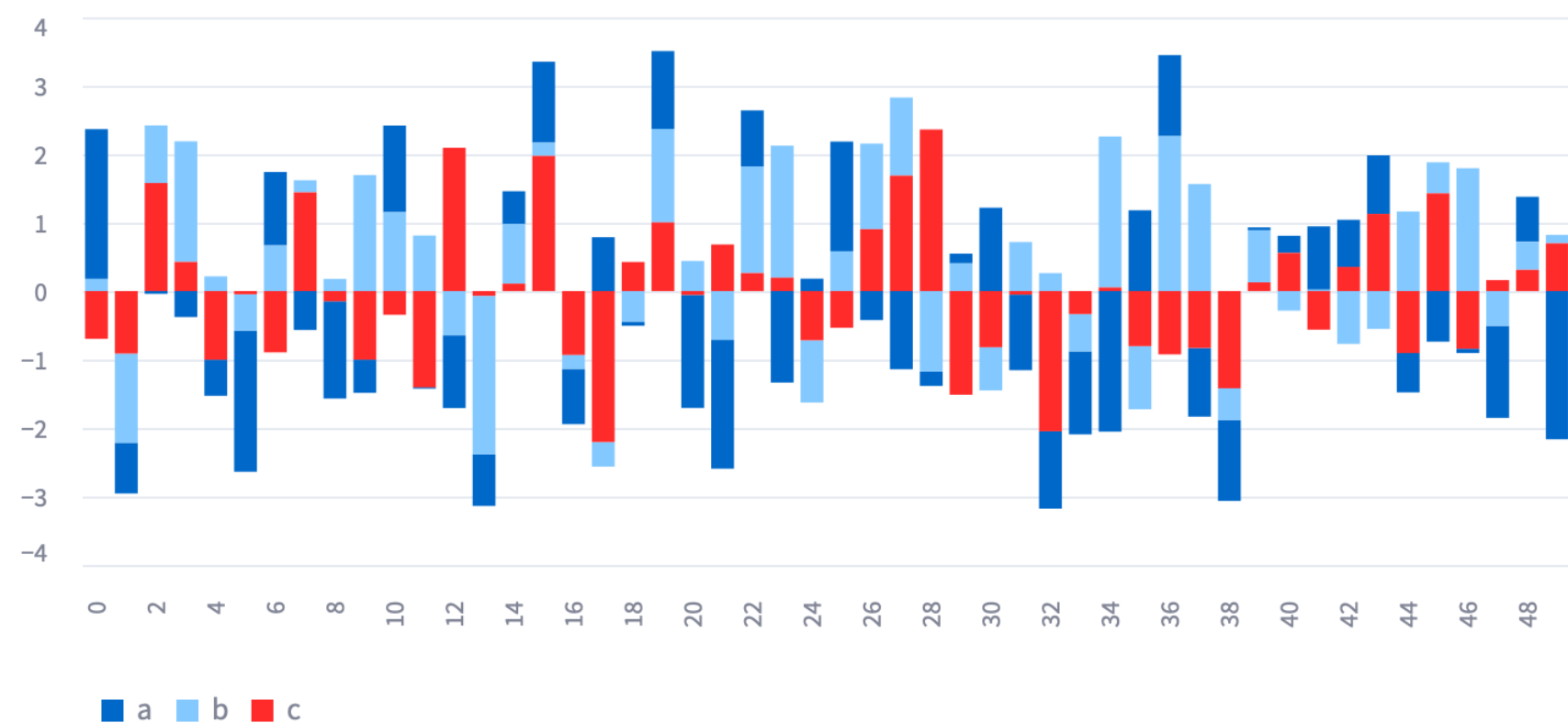
05. Streamlit +
LangChain으로 간단한
웹 어플리케이션 개발

1. 처음 만들어보는
Streamlit 웹 앱 서비스

```
import streamlit as st
import pandas as pd
import numpy as np

chart_data = pd.DataFrame(
    np.random.randn(20, 3),
    columns=["a", "b", "c"])

st.bar_chart(chart_data)
```



막대 그래프를 화면에 출력

데이터의 각 범주가 갖는 양을 손쉽게
비교